Post-Route Static Timing Analysis Report

OpenSTA | sky130_fd_sc_hd | clock: core_clock (period 1.10 ns) | all values read from the post-route report screenshots (actual data)

1. Summary

Metric	Value	Status
Worst setup slack / WNS	-1.762 ns	VIOLATED
TNS (setup)	-69.502 ns	VIOLATED
Worst hold slack	+0.479 ns	MET
Setup clock skew (core_clock)	0.00 ns	OK
Slew / cap / fanout violations	None listed	OK
Total power	7.34 mW	-
Design area	5029 µm² (84% util.)	-

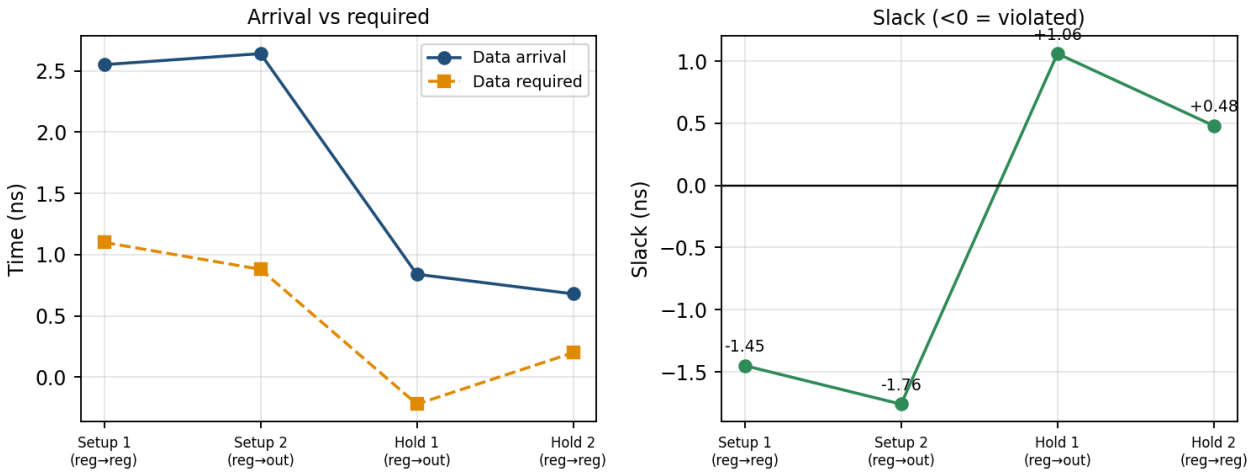
Setup timing fails badly at a 1.10 ns clock, while hold timing passes comfortably. The failing paths all start at the b_reg flip-flops and run through a long chain of xor/majority/xnor gates (an adder-like datapath), which needs roughly 2.5–2.9 ns, far more than the 1.10 ns period.

2. Reported paths

Path	Start → End	Arrival	Required	Slack
Setup 1	b_reg.out[0] → a_reg.out[9] (core_clock)	2.55	1.10	-1.45 (VIOLATED)
Setup 2	b_reg.out[0] → resp_msg[3] (vclk_core_clock)	2.64	0.88	-1.76 (VIOLATED)
Hold 1	ctrl.state.out[1] → resp_val (vclk_core_clock)	0.84	-0.22	+1.06 (MET)
Hold 2	ctrl.state.out[1] → ctrl.state.out[1] (core_clock)	0.68	0.20	+0.48 (MET)

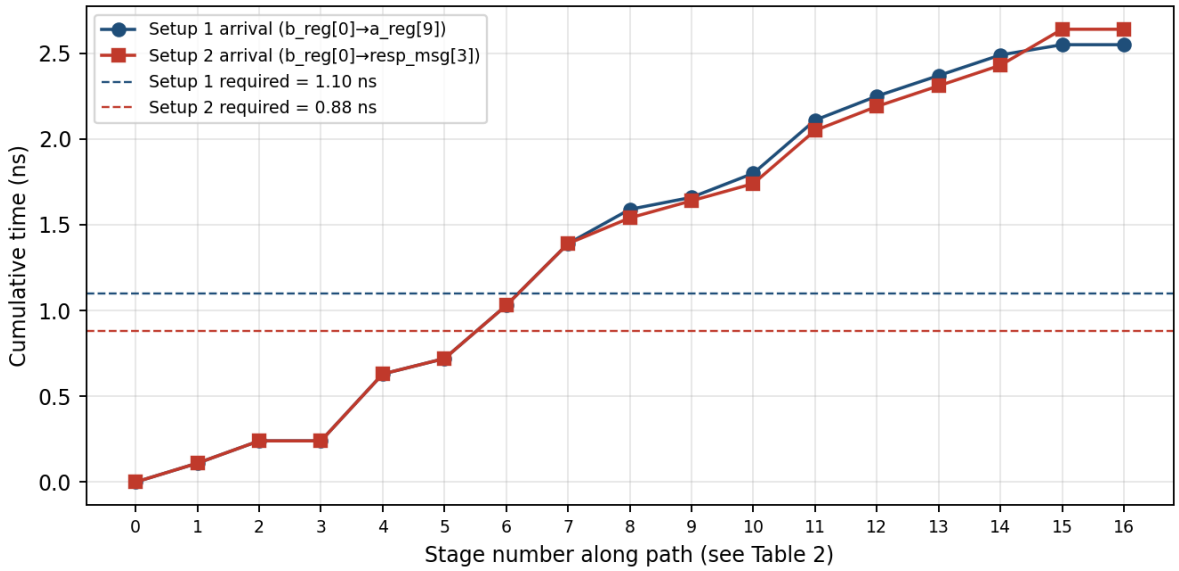
Times in ns. Setup 2 required time = 1.10 (vclk edge) – 0.22 (output external delay) = 0.88. Setup 1 required time = 1.10 (capture clock 1.34 – library setup 0.24).

Fig 3: Setup/hold comparison of the reported paths



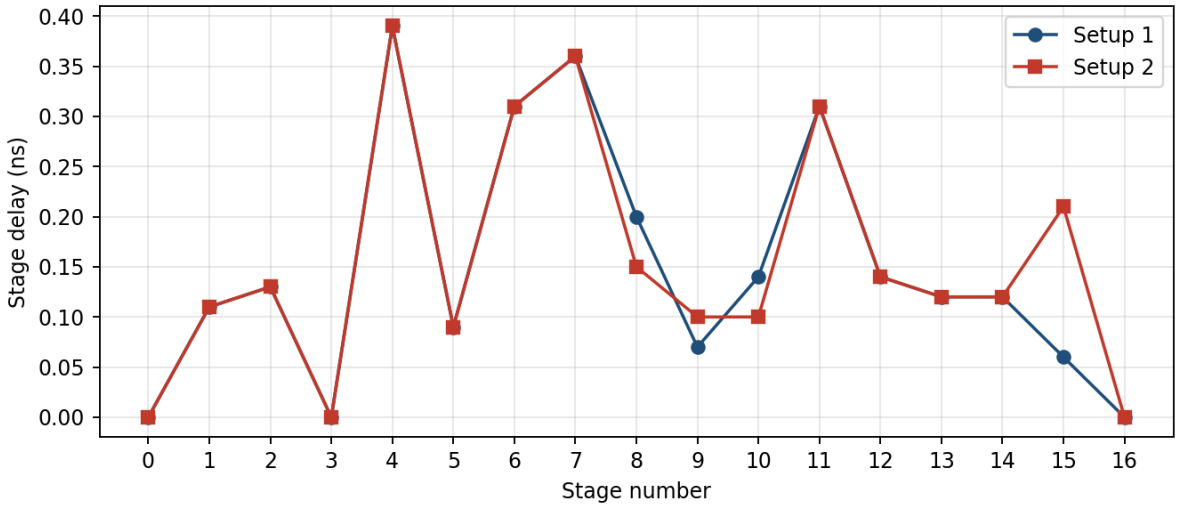
3. Critical path profile

Fig 1: Cumulative data arrival vs required time (setup)



Both paths leave the flip-flop at about 0.63 ns, then climb steadily through the logic. Neither curve comes close to its dashed required-time line; the gap at the end of each line is the slack violation.

Fig 2: Per-stage delay along the two worst setup paths



The biggest single delays are the flip-flop clk→Q (0.38 ns), the maj3 cells (0.31–0.36 ns each) and, on Setup 2, the dlygate4sd2_1 output cell (0.22 ns).

Table 2: Stage-by-stage data (Setup 1 / Setup 2)

#	Setup 1 stage	Delay	Time	Setup 2 stage	Delay	Time
0	clk	0.00	0.00	clk	0.00	0.00
1	clkbuf_0	0.11	0.11	clkbuf_0	0.11	0.11
2	clkbuf_2_0	0.13	0.24	clkbuf_2_0	0.13	0.24
3	FF/CLK	0.00	0.24	FF/CLK	0.00	0.24
4	FF/Q	0.39	0.63	FF/Q	0.39	0.63
5	xor2	0.09	0.72	xor2	0.09	0.72
6	maj3	0.31	1.03	maj3	0.31	1.03
7	maj3	0.36	1.39	maj3	0.36	1.39
8	a21oi	0.20	1.59	buf_4	0.15	1.54
9	o21ai	0.07	1.66	buf_6	0.10	1.64
10	buf_4	0.14	1.80	buf_6	0.10	1.74
11	maj3	0.31	2.11	maj3	0.31	2.05
12	xnor2	0.14	2.25	xnor2	0.14	2.19
13	buf_4	0.12	2.37	buf_4	0.12	2.31
14	mux2i	0.12	2.49	buf_4	0.12	2.43
15	o21ai	0.06	2.55	dlygate	0.21	2.64
16	a_reg/D	0.00	2.55	resp_msg[3]	0.00	2.64

4. Power and area

Group	Internal (W)	Switching (W)	Leakage (W)	Total (W)	%
Sequential	1.65e-03	1.06e-04	3.51e-10	1.75e-03	23.9
Combinational	3.51e-03	1.36e-03	1.87e-09	4.87e-03	66.4
Clock	4.28e-04	2.88e-04	4.77e-11	7.15e-04	9.8
Total	5.58e-03	1.75e-03	2.27e-09	7.34e-03	100

Combinational logic uses two-thirds of the power, matching the deep datapath seen in the timing paths. Area is 5029 µm² at 84% utilization.

5. Conclusions and suggested fixes

• **Setup fails** (WNS -1.762 ns, TNS -69.502 ns) because the logic depth is about 2.3 ns plus clock-to-Q at a 1.10 ns period.
• **Hold passes** (worst +0.479 ns), so no hold buffers are needed; clock skew is 0.00 ns.
• **Estimated feasible period** (derived: 1.10 + 1.762) is about 2.86 ns, roughly 350 MHz, if the design is unchanged.
• **Options:** relax the clock period in the SDC, pipeline the adder-like chain, or reduce the output external delay / replace the dlygate output cell on resp_msg.
• Note: the screenshots show one post-route result; the timing corner/library is not visible in them, so no multi-corner comparison is claimed here.

6. Output screenshots (actual data)

Screenshots of the post-route OpenSTA output used for this report.

Screenshot 1: Setup path 1 – b_reg.out[0] → a_reg.out[9] (part 1)

```
=========== 1. SETUP TIMING (max-delay paths) ===========
Startpoint: dpath.b_reg.out[0]$_DFFE_PP_
            (rising edge-triggered flip-flop clocked by core_clock)
Endpoint: dpath.a_reg.out[9]$_DFFE_PP_
          (rising edge-triggered flip-flop clocked by core_clock)
Path Group: core_clock
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock core_clock (rise edge)
   0.00    0.00   clock source latency
   0.00    0.00 ^ clk (in)
   0.11    0.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_8)
   0.13    0.24 ^ clkbuf_2_0__f_clk/X (sky130_fd_sc_hd__clkbuf_8)
   0.00    0.24 ^ dpath.b_reg.out[0]$_DFFE_PP_/CLK (sky130_fd_sc_hd__edfxtp_1)
   0.38    0.63 ^ dpath.b_reg.out[0]$_DFFE_PP_/Q (sky130_fd_sc_hd__edfxtp_1)
   0.09    0.72 v _120_/_113_/X (sky130_fd_sc_hd__xor2_2)
   0.31    1.03 v _120_/_115_/X (sky130_fd_sc_hd__maj3_2)
   0.36    1.39 v _120_/_117_/X (sky130_fd_sc_hd__maj3_2)
   0.20    1.59 ^ _120_/_119_/Y (sky130_fd_sc_hd__a21oi_4)
   0.07    1.66 v _120_/_120_/Y (sky130_fd_sc_hd__o21ai_2)
   0.14    1.80 v _120_/place165/X (sky130_fd_sc_hd__buf_4)
   0.31    2.11 v _120_/_173_/X (sky130_fd_sc_hd__maj3_2)
   0.14    2.25 v _120_/_174_/Y (sky130_fd_sc_hd__xnor2_2)
   0.12    2.37 v _120_/rebuffer278/X (sky130_fd_sc_hd__buf_4)
   0.11    2.49 ^ _142_/Y (sky130_fd_sc_hd__mux2i_1)
   0.06    2.55 v _144_/Y (sky130_fd_sc_hd__o21ai_1)
   0.00    2.55 v dpath.a_reg.out[9]$_DFFE_PP_/D (sky130_fd_sc_hd__edfxtp_1)
           2.55   data arrival time

   1.10    1.10   clock core_clock (rise edge)
```

Screenshot 2: Setup path 1 – capture side, slack –1.45 (part 2)

```
   0.00    2.55 v dpath.a_reg.out[9]$_DFFE_PP_/D (sky130_fd_sc_hd__edfxtp_1)
           2.55   data arrival time

   1.10    1.10   clock core_clock (rise edge)
   0.00    1.10   clock source latency
   0.00    1.10 ^ clk (in)
   0.11    1.21 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_8)
   0.13    1.34 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_8)
   0.00    1.34 ^ dpath.a_reg.out[9]$_DFFE_PP_/CLK (sky130_fd_sc_hd__edfxtp_1)
   0.00    1.34   clock reconvergence pessimism
  -0.24    1.10   library setup time
           1.10   data required time
---------------------------------------------------------
           1.10   data required time
          -2.55   data arrival time
---------------------------------------------------------
          -1.45   slack (VIOLATED)
```

Screenshot 3: Setup path 2 – b_reg.out[0] → resp_msg[3] (part 1)

```
Startpoint: dpath.b_reg.out[0]$_DFFE_PP_
            (rising edge-triggered flip-flop clocked by core_clock)
Endpoint: resp_msg[3] (output port clocked by vclk_core_clock)
Path Group: vclk_core_clock
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock core_clock (rise edge)
   0.00    0.00   clock source latency
   0.00    0.00 ^ clk (in)
   0.11    0.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_8)
   0.13    0.24 ^ clkbuf_2_0__f_clk/X (sky130_fd_sc_hd__clkbuf_8)
   0.00    0.24 ^ dpath.b_reg.out[0]$_DFFE_PP_/CLK (sky130_fd_sc_hd__edfxtp_1)
   0.38    0.63 ^ dpath.b_reg.out[0]$_DFFE_PP_/Q (sky130_fd_sc_hd__edfxtp_1)
   0.09    0.72 v _120_/_113_/X (sky130_fd_sc_hd__xor2_2)
   0.31    1.03 v _120_/_115_/X (sky130_fd_sc_hd__maj3_2)
   0.36    1.39 v _120_/_117_/X (sky130_fd_sc_hd__maj3_2)
   0.15    1.54 v _120_/rebuffer281/X (sky130_fd_sc_hd__buf_4)
   0.10    1.64 v _120_/rebuffer232/X (sky130_fd_sc_hd__buf_6)
   0.10    1.74 v _120_/place176/X (sky130_fd_sc_hd__buf_6)
   0.31    2.05 v _120_/_162_/X (sky130_fd_sc_hd__maj3_2)
   0.14    2.19 v _120_/_163_/Y (sky130_fd_sc_hd__xnor2_2)
   0.12    2.31 v _120_/place164/X (sky130_fd_sc_hd__buf_4)
   0.11    2.43 v _120_/place74/X (sky130_fd_sc_hd__buf_4)
   0.22    2.64 v output48/X (sky130_fd_sc_hd__dlygate4sd2_1)
   0.00    2.64 v resp_msg[3] (out)
           2.64   data arrival time

   1.10    1.10   clock vclk_core_clock (rise edge)
   0.00    1.10   clock network delay (ideal)
   0.00    1.10   clock reconvergence pessimism
```

Screenshot 4: Setup path 2 – required time, slack –1.76 (part 2)

```
   1.10    1.10   clock vclk_core_clock (rise edge)
   0.00    1.10   clock network delay (ideal)
   0.00    1.10   clock reconvergence pessimism
  -0.22    0.88   output external delay
           0.88   data required time
---------------------------------------------------------
           0.88   data required time
          -2.64   data arrival time
---------------------------------------------------------
          -1.76   slack (VIOLATED)
```

Screenshot 5: Hold path 1 – ctrl.state.out[1] → resp_val, slack +1.06

```
Startpoint: ctrl.state.out[1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by core_clock)
Endpoint: resp_val (output port clocked by vclk_core_clock)
Path Group: vclk_core_clock
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock core_clock (rise edge)
   0.00    0.00   clock source latency
   0.00    0.00 ^ clk (in)
   0.11    0.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_8)
   0.13    0.24 ^ clkbuf_2_0__f_clk/X (sky130_fd_sc_hd__clkbuf_8)
   0.00    0.24 ^ ctrl.state.out[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.30    0.55 v ctrl.state.out[1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.06    0.61 ^ _123_/Y (sky130_fd_sc_hd__clkinv_1)
   0.04    0.65 v _205_/Y (sky130_fd_sc_hd__nor2_4)
   0.10    0.75 v place180/X (sky130_fd_sc_hd__buf_6)
   0.09    0.84 v output55/X (sky130_fd_sc_hd__buf_6)
   0.00    0.84 v resp_val (out)
           0.84   data arrival time

   0.00    0.00   clock vclk_core_clock (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
  -0.22   -0.22   output external delay
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.84   data arrival time
---------------------------------------------------------
           1.06   slack (MET)
```

Screenshot 6: Hold path 2 – ctrl.state.out[1] capture side, slack +0.48

```
 0.00    0.00   clock core_clock (rise edge)
 0.00    0.00   clock source latency
 0.00    0.00 ^ clk (in)
 0.11    0.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_8)
 0.13    0.24 ^ clkbuf_2_0__f_clk/X (sky130_fd_sc_hd__clkbuf_8)
 0.00    0.24 ^ ctrl.state.out[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
 0.00    0.24   clock reconvergence pessimism
-0.04    0.20   library hold time
         0.20   data required time
---------------------------------------------------------
         0.20   data required time
        -0.68   data arrival time
---------------------------------------------------------
         0.48   slack (MET)
```

Screenshot 7: Worst setup/hold slack, WNS, TNS and clock skew

```
=========== 3. WORST SETUP SLACK ===========
worst slack max -1.762

=========== 4. WORST HOLD SLACK ===========
worst slack min 0.479

=========== 5. WNS (setup) ===========
wns max -1.762

=========== 6. TNS (setup) ===========
tns max -69.502

=========== 7. CLOCK SKEW ===========
Clock core_clock
   0.24 source latency ctrl.state.out[0]$_DFF_P_/CLK ^
  -0.24 target latency dpath.a_reg.out[11]$_DFFE_PP_/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew

Clock vclk_core_clock
No launch/capture paths found.
```

Screenshot 8: Slew/cap/fanout check, power and area

```
=========== 8. SLEW / CAP / FANOUT VIOLATORS ===========

=========== 9. POWER ===========
Group                  Internal  Switching    Leakage      Total
                          Power      Power      Power      Power (Watts)
----------------------------------------------------------------
Sequential             1.65e-03   1.06e-04   3.51e-10   1.75e-03  23.9%
Combinational          3.51e-03   1.36e-03   1.87e-09   4.87e-03  66.4%
Clock                  4.28e-04   2.88e-04   4.77e-11   7.15e-04   9.8%
Macro                  0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
Pad                    0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
----------------------------------------------------------------
Total                  5.58e-03   1.75e-03   2.27e-09   7.34e-03 100.0%
                          76.1%      23.9%       0.0%

=========== 10. AREA ===========
Design area 5029 um^2 84% utilization.

Post-route STA report commands completed.
```